**A**

**Mini Project Report**

**on**

# <u>DoRev: AI-Powered Google Review Manager</u>

Submitted in partial fulfillment of the requirements for the degree

**Third Year Engineering – Computer Science and Engineering Data Science**

**by**


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# CERTIFICATE

This to certify that the Mini Project report on **DOREV: AI-POWERED GOOGLE REVIEW MANAGER** has been submitted by Asma Rajguru (23207005), Anas Chougle (23207006), Arju Salmani (22107003), Yash Nalawade (22107058) who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **ComputerScience Engineering (Data Science)**, during the academic year **2024-2025** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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# ABSTRACT

DoRev is an AI-powered Google review management system designed to redefine how businesses handle and respond to customer feedback. By integrating advanced technologies such as Sentiment Analysis, Natural Language Processing (NLP), and a Toxicity Classification Filter, the platform automates the review handling process and enhances the quality of visible feedback. It intelligently analyzes customer sentiments, filters out negative or harmful reviews, and highlights constructive ones—improving review relevance and boosting public perception. Developed using ReactJS, Tailwind CSS, TypeScript, and NLP models, DoRev combines a powerful backend with an intuitive frontend to deliver a seamless and efficient experience.

A key innovation of DoRev is its toxicity detection feature, which protects businesses from reputational damage by automatically identifying and suppressing toxic or misleading content. To further support business growth, the system features a dynamic dashboard that visualizes sentiment trends, customer feedback insights, and satisfaction levels—allowing for strategic, data-driven decisions. DoRev addresses major pain points in manual review handling, such as inefficiency, poor analysis, and inconsistent review quality. With its QR-code-based review submission, businesses can gather real-time feedback and act promptly. By blending AI-driven sentiment classification, toxicity filtering, and feedback analytics, DoRev represents a major leap in digital reputation management, offering a smarter, faster, and more reliable way to manage online reviews.

# CHAPTER 1

## Introduction

Finding the right tools to manage customer reviews can be challenging due to the inefficiencies in manual processes and inaccurate sentiment analysis in existing systems. DoRev: AI-Powered Google Review Manager is an automated review management system that uses advanced Sentiment Analysis and Natural Language Processing (NLP) to categorize and filter reviews based on their relevance. The platform helps businesses highlight positive and constructive feedback while filtering out harmful or irrelevant content using a Toxicity Classification Filter. Built with ReactJS, Tailwind CSS, and TypeScript, DoRev provides a dynamic dashboard for real-time insights into customer feedback, enabling businesses to make data-driven decisions and maintain a strong online reputation with minimal effort.

### 1.1 Purpose:

The primary purpose of DoRev: AI-Powered Google Review Manager is to streamline the process of managing customer feedback and protecting brand reputation. Traditional review management systems often rely on manual sorting and fail to offer meaningful insights. DoRev addresses these challenges by using advanced AI technologies, such as Sentiment Analysis and Toxicity Classification, to automate review categorization and analysis, ensuring businesses can focus on constructive feedback while safeguarding their reputation.

**Efficient Review Management:**

DoRev simplifies Google review management by automating the analysis and categorization of feedback. Through Sentiment Analysis, the system identifies the tone of each review, ensuring only relevant and positive feedback is highlighted. Continuous refinement of algorithms ensures businesses receive the most valuable insights from customer reviews.

**Brand Reputation Protection:**

Many traditional systems fail to filter out harmful reviews effectively. DoRev helps businesses protect their brand by automatically filtering out negative content and showcasing only constructive reviews, ensuring the brand maintains a strong and trustworthy online presence.

**Data-Driven Insights:**

Leveraging NLP and Sentiment Analysis, DoRev enables businesses to gain a deeper understanding of customer sentiments and trends. The platform's dynamic dashboard visualizes valuable feedback insights, helping businesses make informed decisions to improve customer satisfaction and reputation over time.

**Motivation:**

Handling customer reviews can be a time-consuming and error-prone task, especially when businesses are overwhelmed with large volumes of feedback and negative comments. DoRev: AI-Powered Google Review Manager addresses these challenges by automating the analysis and filtering of reviews. By leveraging advanced AI technologies, DoRev ensures businesses can easily highlight valuable feedback, protect their brand reputation, and gain actionable insights, all while reducing the time and effort spent on manual review management.

## 1.2 Problem Statement:

The process of managing and analyzing customer reviews can be overwhelming and inefficient due to the sheer volume of feedback and the inability to extract meaningful insights in real time. Many traditional review systems fail to automate the filtering and analysis of feedback, leading to businesses struggling with irrelevant or harmful reviews.

**Challenges Faced by Businesses:**

- Manual Review Handling: Businesses spend significant time sorting through large amounts of feedback manually, leading to inefficiencies.
- Inaccurate Sentiment Analysis: Traditional systems often fail to accurately interpret the tone and content of reviews, causing missed opportunities.
- Toxic Reviews: Negative or toxic reviews often remain visible, harming a business's reputation.

**Solutions:**

DoRev tackles these challenges by using advanced AI technologies, including Sentiment Analysis and Toxicity Classification filters, to automate review management. The system ensures businesses only showcase high-quality feedback by filtering out irrelevant or harmful comments. Additionally, DoRev provides businesses with actionable insights through real-time analytics, helping them improve customer engagement and protect their online reputation. With its intuitive interface and data-driven approach, DoRev simplifies review management and brand reputation management.

## 1.3 Objectives:

Managing online reviews is a critical part of maintaining a business's reputation, but the process can be time-consuming and inefficient. DoRev simplifies this by using advanced AI-powered sentiment analysis and toxicity detection to automate review management. Customers submit feedback by scanning a QR code, and the system instantly analyzes, filters, and categorizes the reviews. By surfacing constructive feedback and blocking harmful content, DoRev helps businesses gain valuable insights, improve customer trust, and maintain a strong online presence with minimal effort.

- Improve Review Management Efficiency: DoRev automates the collection, analysis, and classification of Google reviews using Sentiment Analysis AI. This reduces manual effort and helps businesses process large volumes of feedback quickly, leading to faster and more informed decision-making.

- Enhance Review Quality: By filtering out irrelevant, misleading, or spammy reviews, DoRev ensures that only positive and constructive feedback is highlighted. This maintains a clean and trustworthy review section, strengthening customer confidence and brand perception. restaurant.

- Facilitate Insightful Analysis: An interactive dashboard displays visual insights like customer sentiment trends, keywords, and recurring feedback themes. This helps businesses understand customer needs better and make data-driven improvements. experience.

- Protect Brand Reputation: With a built-in toxicity classification filter, DoRev automatically detects and removes harmful or inappropriate reviews. This protects businesses from negative exposure and helps maintain a credible online presence.

### 1.4 Scope:

The scope of DoRev defines the boundaries of the project, focusing on automating review management, improving sentiment analysis accuracy, and ensuring businesses can maintain a strong brand reputation with minimal effort. The system integrates AI technologies for efficient review categorization and provides actionable insights to enhance customer engagement.

**Technical Scope:**

- **Machine Learning Algorithms:** DoRev utilizes Sentiment Analysis and Toxicity Classification to filter and analyze customer reviews. The system evaluates tone and content of feedback to categorize reviews as positive, negative, or neutral, ensuring that only high-quality feedback is showcased. NLP algorithms enhance review filtering accuracy and insight generation.

- **Backend Development:** The backend of DoRev is developed using TypeScript to handle complex data processing, sentiment analysis, and integration of AI models. This approach ensures efficient review management, providing them with valuable feedback insights.

- **Frontend Development:** The user interface of DoRev is built using ReactJS and Tailwind CSS, providing a clean, responsive, and intuitive experience for users. The platform enables businesses to easily manage reviews, track performance, and access insights, all in one place.

- **Database Management:** The system uses cloud-based databases to store review data, user feedback, sentiment analysis results, and brand reputation metrics. This ensures secure and scalable data handling, supporting integration and review management for businesses.

**Functional Scope:**

- **Review Filtering:** Uses Sentiment Analysis to categorize reviews, ensuring that only positive and constructive feedback is displayed.
- **Sentiment Insights Generation:** Implements NLP to analyze reviews and provide valuable insights for businesses to improve customer engagement.
- **Toxicity Detection:** Automatically filters out negative or toxic reviews, protecting the business's reputation.
- **Analytics Dashboard:** Provides a real-time dashboard to visualize trends and key metrics, aiding businesses in decision-making.

# CHAPTER 2

## Literature Review

DoRev project highlights recent advancements in AI-powered review management systems, focusing on Sentiment Analysis, Toxicity Detection, and Natural Language Processing (NLP). These studies demonstrate innovative approaches to improving review quality, automating moderation, and protecting brand reputation, ensuring businesses receive accurate and constructive feedback. By analyzing these research contributions, we aim to integrate best practices into DoRev, building a smarter and more reliable platform for managing Google reviews efficiently.

In the study ″Modelling the Impact of Hotel Facilities on Online Hotel Review Score for City of Skopje″ (2021)[1] by Marija Stojchevska, Andreja Naumoski, and Kosta Mitreski, the authors analyzed how various hotel facilities influence online review scores in Skopje. Using data from 80 hotels listed on Booking.com and TripAdvisor, and employing ArcGIS software for spatial analysis, the study examined the impact of 20 different hotel facilities on customer reviews. The findings revealed that certain amenities, particularly soundproofing and free parking, have a significant positive effect on review scores, with this impact being especially notable in 3 and 4-star hotels.

Another notable study, "Restaurant Reviews Analysis Model Based on Machine Learning Algorithms" [2] by Yan Jiang (2022), presents an advanced machine learning-based sentiment analysis system for evaluating restaurant reviews. The research integrates Supervised Learning, Unsupervised Learning, and Reinforcement Learning techniques to classify customer sentiments effectively. The methodology includes text preprocessing, feature extraction using TF-IDF (Term Frequency-Inverse Document Frequency), and classification using models like Naïve Bayes and SVM. The study highlights the importance of review-based sentiment analysis in restaurant recommendation systems, as it provides unbiased ratings and better decision-making insights for both customers and businesses. The use of sentiment analysis to filter misleading or biased reviews improves the accuracy of restaurant recommendations, ensuring reliable user experiences.

In the 2023 study ″Word Cloud of Online Hotel Reviews in Myanmar for Customer Satisfaction Analysis″ [3]by Phasit Charoenkwan, the author explored the key factors contributing to customer satisfaction and dissatisfaction in the Myanmar hotel industry. By analyzing 682 online reviews from TripAdvisor using word cloud visualization and clustering techniques, the study identified prominent themes in guest experiences. Positive reviews frequently mentioned aspects such as room quality, staff, pool, view, and food, while negative reviews commonly highlighted issues with room cleanliness, internet connectivity, and staff performance.

# CHAPTER 3

## Proposed System

The proposed system, DoRev: AI-Powered Google Review Manager, is an AI-driven platform that automates the management of customer feedback to enhance brand reputation. It utilizes Sentiment Analysis and Toxicity Classification filters to analyze and categorize Google reviews, ensuring businesses focus on high-quality feedback while filtering out harmful content. The system features a frontend developed with ReactJS and Tailwind CSS for a user-friendly interface, while the backend, built with TypeScript, integrates AI algorithms to provide real-time insights and streamline review management.

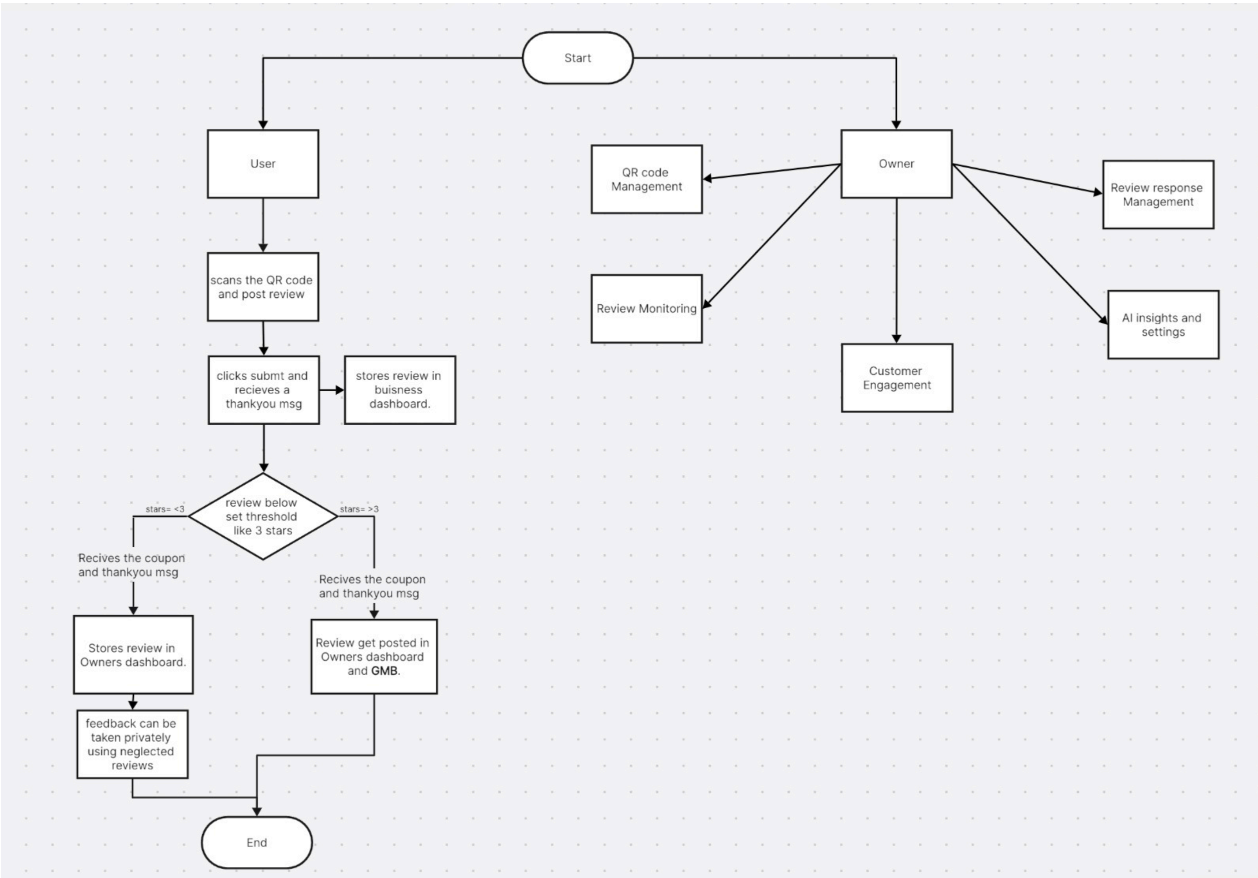



Figure 3.1 Flow Chart

In the Figure 3.1, The system interface offers a user-centric experience designed to enhance review management through four key modules, as illustrated in the following sections. Each module is tailored to streamline processes and provide valuable insights based on user inputs and integrated data.

- Review Management Module: This module allows businesses to automatically categorize and analyze customer feedback. Reviews are processed using Sentiment Analysis, and only positive and constructive feedback is displayed for further actions. This module is designed to simplify the review management process by ensuring businesses focus on relevant, high-quality feedback.

- Sentiment Analysis Module: Leveraging Natural Language Processing (NLP), this module analyzes the tone of each review and categorizes it accordingly. The interface provides a visual representation of sentiment trends, such as bar graphs or pie charts, enabling businesses to track customer opinions and adjust their strategies based on customer feedback.

- Toxicity Tracking Module: This module integrates a toxicity filter to identify and remove harmful or inappropriate reviews. By automatically flagging toxic content, businesses can protect their brand reputation from harmful reviews, ensuring that only valuable, constructive feedback is visible to potential customers.

- Analytics Dashboard Module: The dashboard provides a comprehensive view of customer feedback, tracking metrics such as sentiment trends, review volume, and overall reputation. The module uses visual data to offer insights that help businesses make data-driven decisions, improve customer satisfaction, and enhance brand perception over time.

### 3.1 Features and Functionality

The DoRev platform is designed to streamline review management for businesses using AI-powered insights and real-time data analysis. With a focus on improving brand reputation and decision-making, the system integrates advanced features to ensure businesses receive relevant, high-quality feedback while maintaining a strong online presence.

- **User-Friendly Interface:**

The platform offers an intuitive interface that allows businesses to easily manage their reviews, automatically categorizing them based on sentiment and relevance. Users can access actionable insights and track customer feedback effortlessly.

- **Automated Review Management:**

DoRev uses AI-powered Sentiment Analysis to classify reviews, ensuring businesses can focus on positive, constructive feedback while ignoring irrelevant or harmful comments. This reduces manual effort and improves overall efficiency in review handling.

- **Advanced Machine Learning Algorithms:**

The system employs Natural Language Processing (NLP) and machine learning algorithms to analyze customer feedback. This ensures accurate categorization of reviews, identifying sentiments like positive, neutral, and negative, which helps businesses respond more effectively.

- **Toxicity Detection for Reputation Protection:**

DoRev integrates a toxicity classification filter to detect harmful, spam, or inappropriate reviews. By automatically filtering out these reviews, the platform protects businesses' reputations from misleading or malicious feedback.

- **Analytics Dashboard:**

The platform features a comprehensive dashboard that visualizes review trends, customer sentiment, and overall feedback performance.

# Chapter 4

# Requirement Analysis

Requirement analysis is a crucial phase in software development, ensuring that the system meets the specific needs of businesses while providing effective review management and reputation protection. This phase outlines the scope, functionality, and technology stack required to build a platform that automates review categorization and analysis, offering valuable insights while safeguarding brand reputation. It involves identifying key stakeholders, understanding system requirements, defining functional and non-functional needs, designing a robust database schema, and selecting the best technologies to ensure a seamless user experience.

The functional requirements of DoRev include automated review classification through Sentiment Analysis, Toxicity Classification to filter harmful feedback, real-time review analytics, and a dynamic dashboard for actionable insights. These features help businesses manage customer feedback efficiently by ensuring only meaningful and positive reviews are highlighted, reducing the impact of negative or irrelevant comments. The system also must support user authentication, allowing businesses to securely manage their profiles and review data. Additionally, the platform must be capable of storing and managing large volumes of reviews, sentiment data, and business information within a well-structured database. Beyond core functionalities, DoRev must adhere to key non-functional requirements such as high performance, scalability, security, and user accessibility. It must handle large amounts of review data, providing fast and reliable responses, while ensuring that all data is secure and protected from unauthorized access.

To meet these requirements, DoRev uses a structured database schema that includes entities like Users, Reviews, Sentiment Data, and Toxicity Classifications. The technology stack comprises ReactJS and Tailwind CSS for the frontend, TypeScript for backend operations, and MySQL for database management. The system integrates NLP-based Sentiment Analysis to classify review sentiments and a toxicity filter to remove harmful content. Additionally, the platform leverages AI to provide insightful analytics, helping businesses make data-driven decisions and protect their brand reputation.

# Chapter 5
# Project Design

The project design of DoRev: AI-Powered Google Review Manager acts as a strategic framework for creating an advanced system that automates review management and enhances brand reputation. Focusing on a user-friendly and efficient interface, the platform integrates powerful AI techniques like Sentiment Analysis and Toxicity Classification for intelligent feedback processing. The design ensures high performance, scalability, and security while utilizing a well-structured database and a robust technology stack. By incorporating real-time review analytics and AI-powered insights, DoRev addresses the challenges businesses face in managing customer feedback, setting the stage for continuous growth and innovation in brand reputation management.

## 5.1 Use Case Diagram:

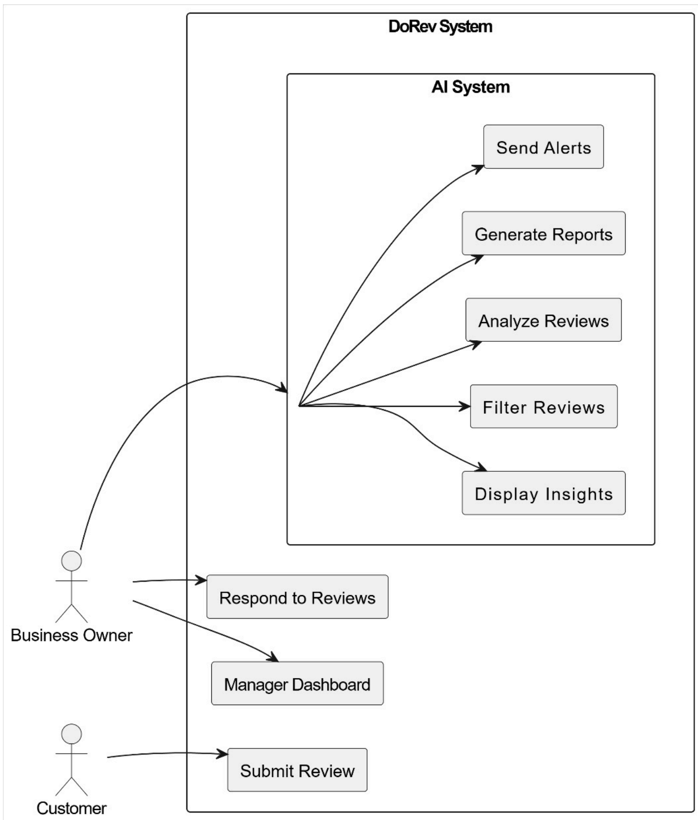



Figure 5.1.1 Use Case Diagram

In the Figure 5.1.1, The flowchart represents the DoRev system, which provides automated review management, sentiment analysis, toxicity filtering, and feedback insights.

- Actors:

The primary actors are Business Owner, Customer, and AI System, each represented by a stick figure.

- System:

The central element is the "AI-Powered Google Review Manager," depicted as a large box. This represents the core functionality of the DoRev system.

- Use Cases:

The use cases are represented by ovals connected to the AI-Powered Google Review Manager. They represent specific interactions between users and the system.
The use cases in the diagram are:

1. Submit Review: This use case represents the process of submitting a review, typically initiated by a customer.

2. Respond to Reviews: This use case represents the business owner's ability to view and reply to customer feedback.

3. Manage Dashboard: This use case represents the ability to monitor trends, insights, and satisfaction scores through a visual dashboard.

4. Analyze Review: This use case represents the use of sentiment analysis and NLP to understand the tone and content of reviews.

5. Filter Reviews: This use case represents the process of detecting and hiding toxic or misleading reviews using the toxicity classification filter.

6. Display Insights: This use case represents the generation and display of actionable insights and trends from the review data.

7. Generate Reports: This use case represents the creation of analytical reports that summarize performance over time.

8. Send Alerts: This use case represents the notification system that informs business owners of urgent feedback or sentiment shifts.

- **Relationships:**

Lines connecting the actors, system, and use cases represent relationships between them. Solid lines indicate direct interactions, while labeled arrows indicate internal system processes. For example, the customer interacts directly with the “Submit Review” use case, while the AI System handles internal processes like analysis and filtering.

- **External Entities:**

The Sentiment Analysis Model, Toxicity Classifier, and Database represent external components that interact with the system. The Sentiment Analysis Model and Toxicity Classifier provide AI capabilities for understanding and filtering reviews. The Database stores user data, reviews, and reports essential for analysis and dashboard display.

Overall, the diagram illustrates how users interact with the DoRev system to manage reviews, extract insights, and maintain online reputation efficiently.

### 5.2 DFD (Data Flow Diagram):

The Fig 5.2.1 Data Flow Diagram (DFD) for the DoRev (AI-Powered Google Review Management System) illustrates the systematic movement of data between users, system processes, external modules, and data stores.

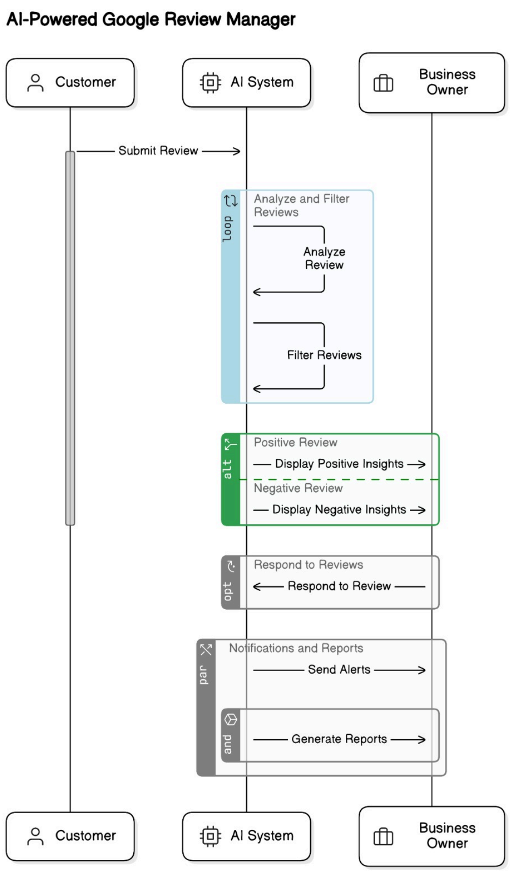



Figure 5.2.1 Data Flow Diagram

- **User Interaction:**

Users interact with the system through various points:

1. Submit Reviews: Customers initiate the process by submitting feedback through the review interface or QR-based submission.
2. Respond to Reviews: Business owners can directly engage with submitted reviews by providing public replies.
3. Access Dashboard Insights: Business owners can view sentiment trends, feedback summaries, and performance reports via the dashboard.
4. Receive Alerts: Users, especially business owners, receive alerts on toxic reviews or significant sentiment changes.

- **System Processes:**

1. Sentiment Analysis Engine: This core process uses NLP to understand the emotional tone and intent behind each submitted review, enabling intelligent categorization.
2. Toxicity Filtering Module: This process classifies and filters out inappropriate, harmful, or misleading reviews to maintain quality and protect reputation.
3. Insight Generation Engine: This process transforms analyzed data into visual insights, trends, and analytics for business owners to act upon.

- **External Systems:**

1. Google Reviews API: This API is used to pull and post reviews in real-time, ensuring up-to-date feedback availability and synchronization.
2. AI/NLP Models: These are pre-trained and custom models used for sentiment detection and toxicity classification to power intelligent decision-making.

- **Data Stores:**

1. Customer Feedback Database:
   Stores all submitted customer reviews along with metadata like timestamps and processed sentiment results for future analysis.

2. Toxic Review Logs:
   Maintains a record of flagged or suppressed reviews identified as toxic or misleading for moderation and auditing.

3. Dashboard Data Store:
   Holds sentiment trends, metrics, and visualized insights displayed on the business owner's dashboard.

4. Alert Log Repository:
   Stores alert details including timestamps, severity, and review context for tracking and quick action

- **Data Flow:**

Data flows cyclically between users, processes, and data stores. Customer reviews are submitted and stored, then passed to the NLP engine for sentiment analysis. The toxicity module filters harmful content, and the insight engine processes clean data into actionable insights. Alerts are triggered based on system thresholds. Business owners interact with reports and respond to reviews. System feedback loops improve accuracy and performance over time.

In summary, the DFD illustrates how user submissions, AI-driven processes, integrated APIs, and structured data stores work together to deliver real-time review management, automated moderation, and actionable insights through the DoRev platform.

### 5.3 System Architecture:

The Fig 5.3.1 illustrates how users, the AI system, and business owners interact within the DoRev platform and how its main components work together. The process begins when a user submits a review, which is then processed by the AI engine. The system filters, analyzes, and stores reviews, while business owners receive alerts, reports, and sentiment insights through a visual dashboard.

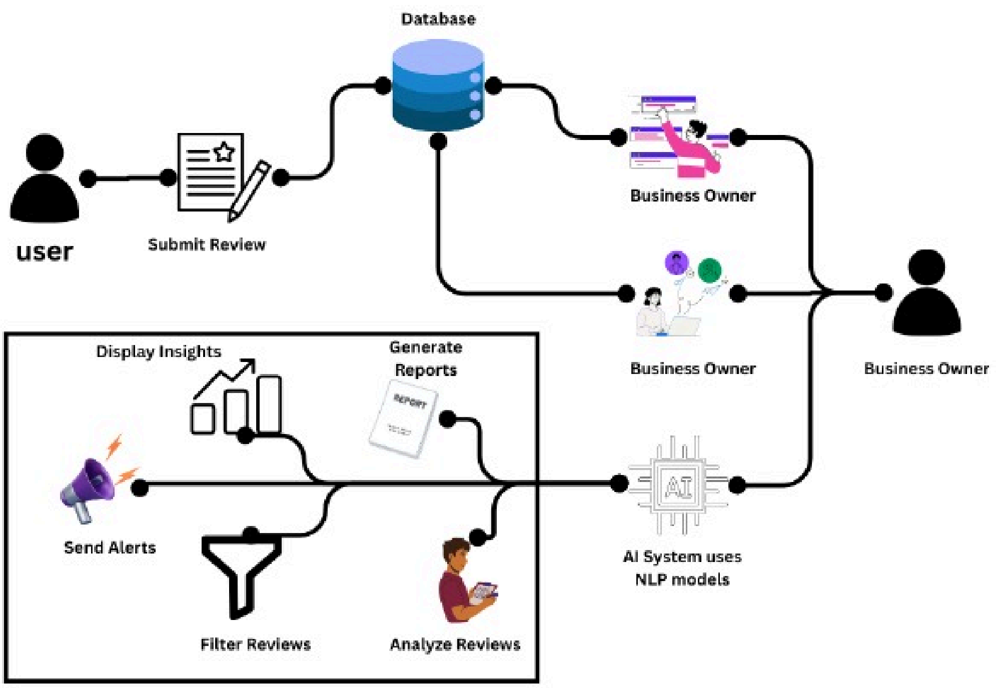



Figure 5.3.1 System Architecture

- Restaurant Recommendation: Users interact with the system by providing their preferences. The system uses this information along with data from the "DATASET" to suggest suitable restaurants.
- Review Analysis: The AI system utilizes advanced NLP models to analyze submitted reviews. It evaluates sentiment, extracts keywords, and detects toxicity. This processing ensures the relevance and safety of displayed feedback.
- Sentiment Insights: Based on analysis, the system provides sentiment-driven insights. These are then sent to the business owner via a reporting module, offering an overview of positive and negative trends.

- Alerts and Reports: When harmful or highly emotional content is detected, the system triggers alerts. Simultaneously, periodic reports are generated to help business owners monitor customer satisfaction and review trends.
- A centralized database supports the platform, storing raw reviews, insights, toxicity flags, and report data. This ensures smooth data access for real-time decision-making and dashboard rendering.

The architecture effectively integrates user reviews, AI processing, database operations, and business-facing insights to streamline digital reputation management.

## 5.4 Implementation:

The implementation phase of the project involves transforming the conceptual framework into a fully functional AI-driven review management system. Screenshots of the user interface (UI) at various stages visually demonstrate the development and user experience aspects of the platform. Each screenshot is paired with key insights regarding the page's role, core functionalities, and integrated features such as sentiment filtering, dashboard access, and review submission options. These visuals and annotations document the development lifecycle, confirming that the end product aligns with the intended goals and user-centric design expectations.

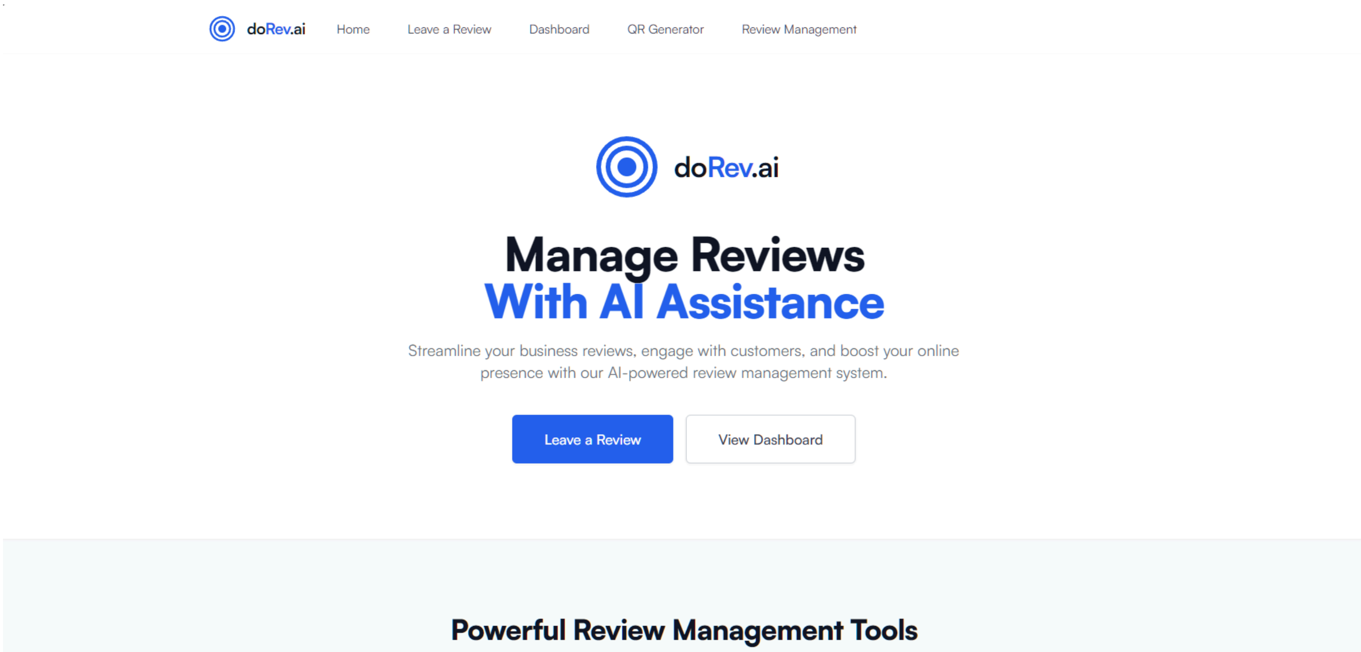



Figure 5.4.1 Sign In Page

The Figure 5.4.1 is the landing page of the DoRev platform and features a clean, intuitive layout. At its center, users are presented with two prominent buttons—'Leave a Review' and 'View Dashboard'—for easy navigation. Above this, a headline introduces the purpose of the platform, highlighting AI-powered review management. The page includes a minimalistic design with a centered logo and clear call-to-action elements, providing a seamless and professional first impression.

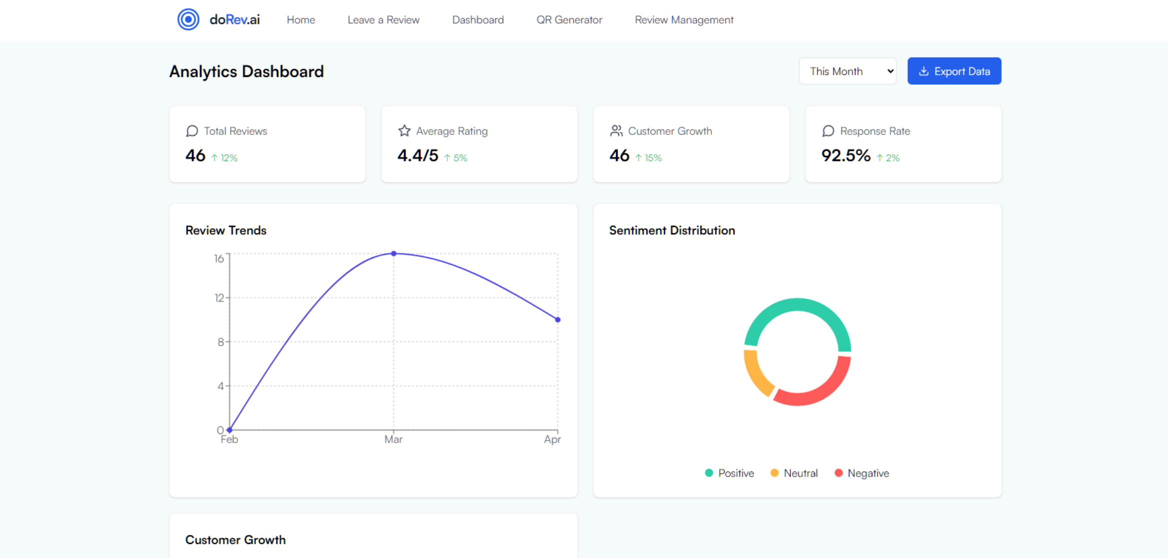



Figure 5.4.2 Dashboard Page

The Fig 5.4.2 is the dashboard page of the DoRev platform and offers a visually informative, easy-to-navigate interface with key metrics clearly displayed for quick insights. At the top, users can view performance indicators like "Total Reviews," "Average Rating," "Customer Growth," and "Response Rate," all presented through well-defined data cards. Each section is supported by graphs and charts, such as “Review Trends” and “Sentiment Distribution,” providing visual representation of customer feedback over time. The dashboard is designed with a modern, clean layout, enabling businesses to monitor analytics efficiently and make data-driven decisions with ease. The arrangement ensures all insights are easily accessible, supporting streamlined review management.

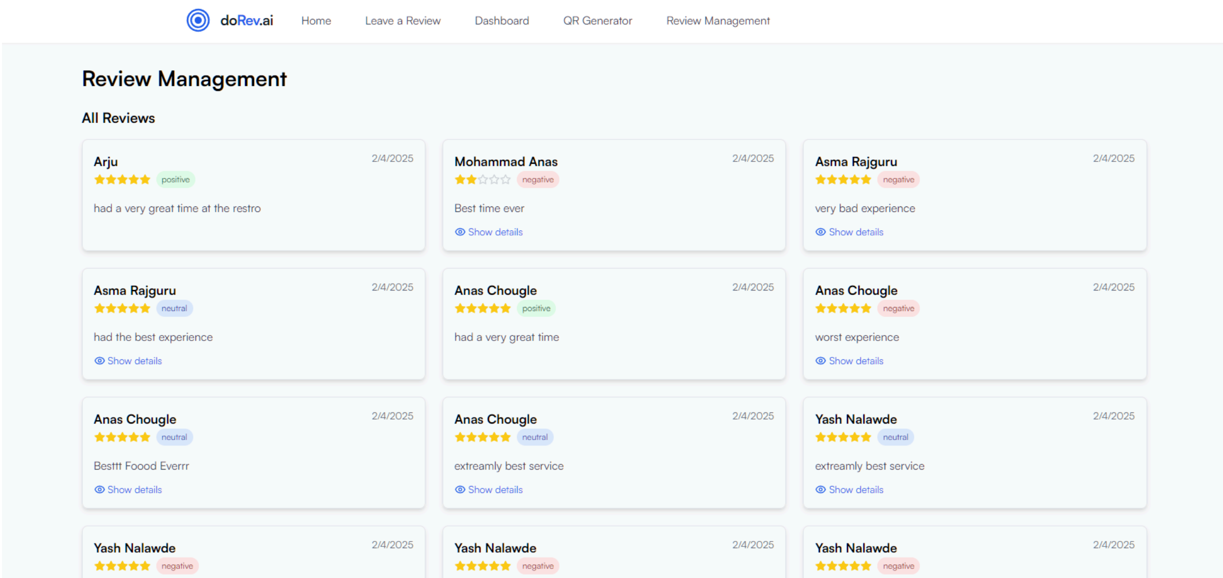



*Figure 5.4.3 Personalized Recommendations Page*

The Fig 5.4.3 is the Review Management page on the doRev.ai platform that displays a structured and user-friendly interface to manage and monitor customer feedback efficiently..

The layout includes customer reviews presented in card format, each displaying the reviewer's name, date, star rating, sentiment tag (positive, neutral, or negative), and review content. Beneath each review is a "Show details" link for accessing full review data. The page is cleanly designed for quick readability, enabling admins to evaluate and act on user feedback with clarity. This layout ensures all reviews are organized and accessible, supporting a streamlined review management process.

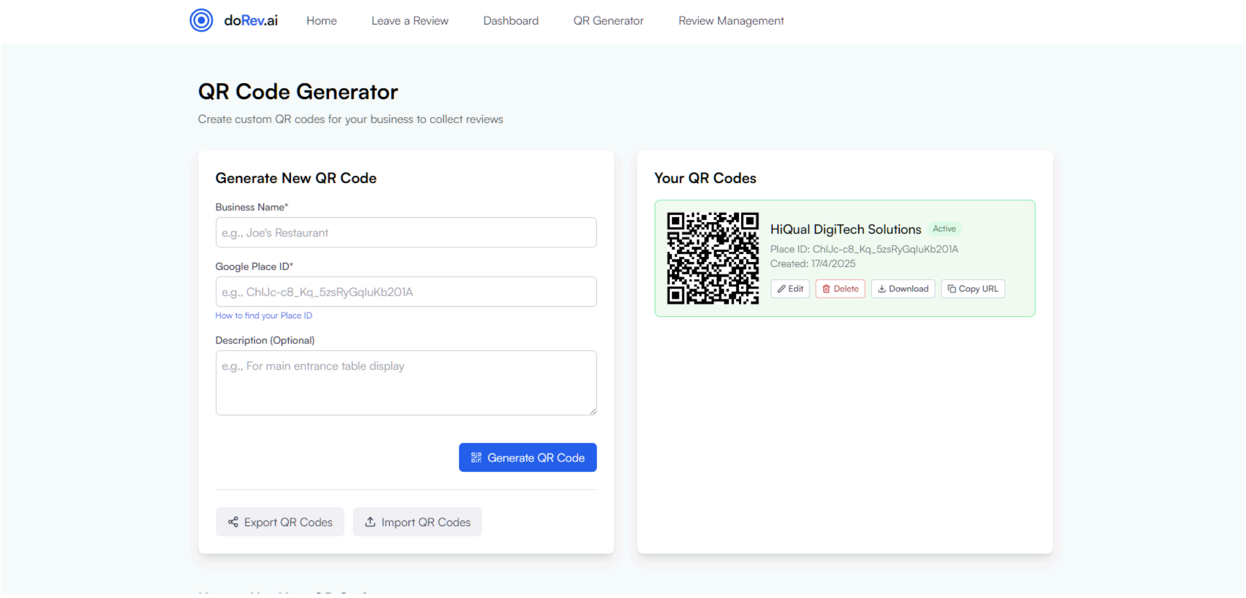



Figure 5.4.4 QR Code Generator page

The Fig 5.4.4 is the QR Code Generator page on the doRev.ai platform that presents users with an interactive form interface to generate and manage review collection QR codes. Users can enter a "Business Name" and the corresponding "Google Place ID" using the text input fields on the left panel. The form includes an optional description box to specify QR placement, such as for display at tables or counters. Once submitted, the generated QR code appears on the right side, showing the business name, place ID, and creation date. Management options like Edit, Delete, Download, and Copy URL are provided. The interface is designed with a clean and functional layout, allowing businesses to quickly create and deploy custom QR codes with ease.

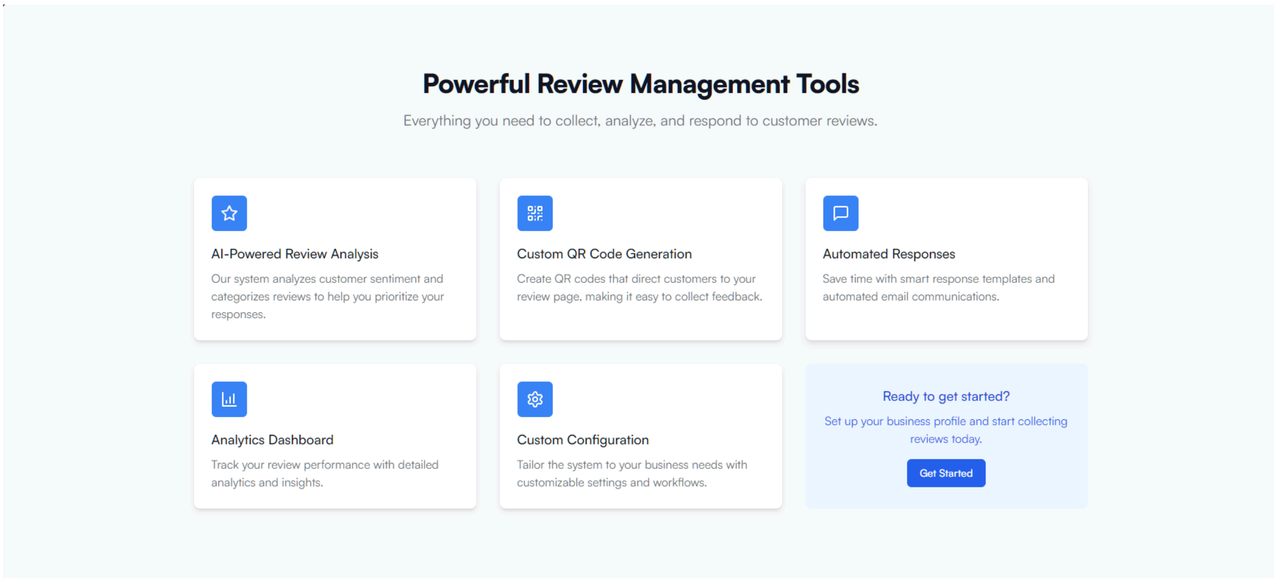



Figure 5.4.5 Powerful Review Management Tools section

Fig 5.4.5 is the "Review Management" page of the doRev.ai platform. It presents users with a structured and user-friendly grid showcasing the key features for efficient customer feedback handling. The section includes blocks for "AI-Powered Review Analysis," "Custom QR Code Generation," and "Automated Responses."Each feature card is clearly labeled and briefly described to help users understand its functionality. The layout is clean and organized, with a prominent "Get Started" panel guiding businesses to initiate their review setup. The interface ensures ease of navigation while offering powerful capabilities.

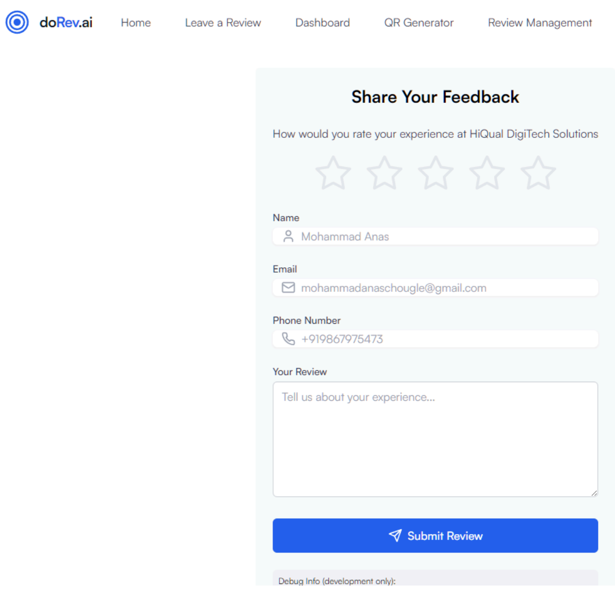


Figure 5.4.6 Review Submission page

Fig 5.4.6 is the Review Submission page of the doRev.ai platform. It presents users with an interface to share their experience directly with the business. At the top, users can rate their visit using a 5-star scale, helping capture feedback sentiment visually.Below the rating section, users are prompted to enter their name, email, phone number, and write a detailed review in a text box. The page concludes with a prominent "Submit Review" button to finalize the submission. The layout is simple and intuitive, ensuring a seamless user experience while collecting essential feedback.

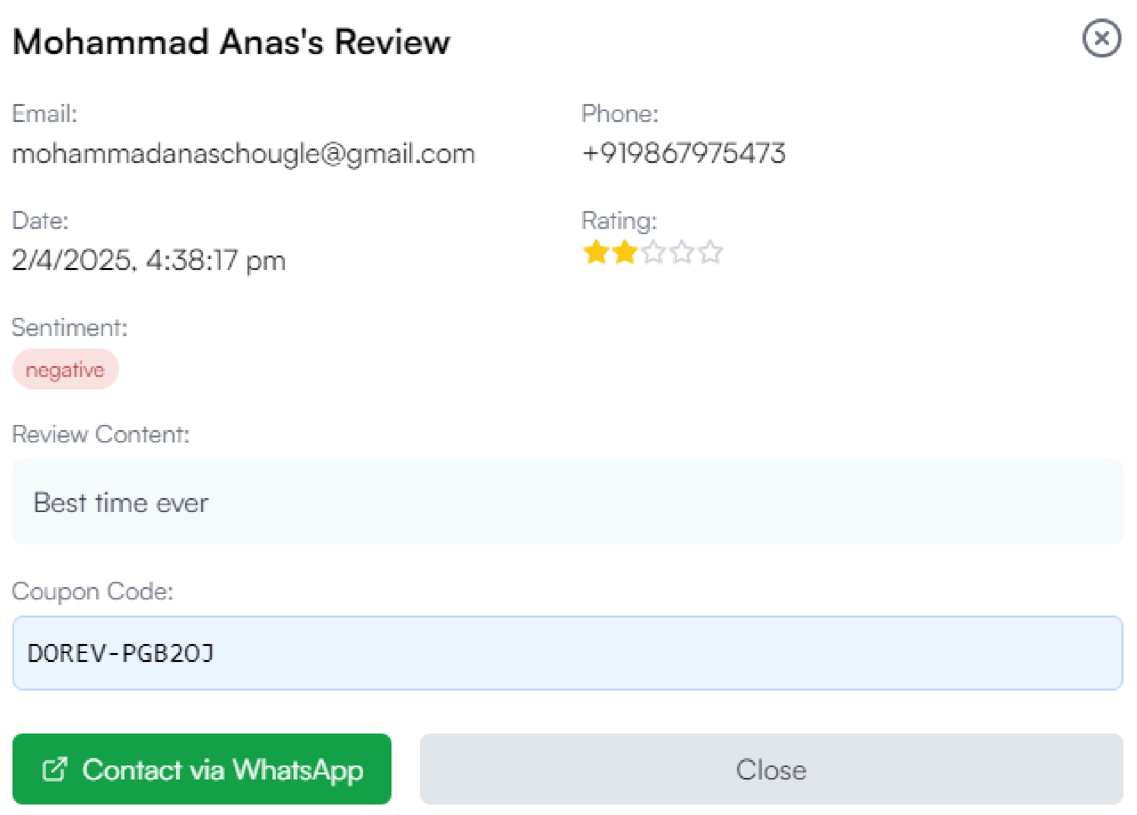



Figure 5.4.7 Review Details Popup page

Fig 5.4.7 is the Review Details popup of the doRev.ai platform. It displays a user’s submitted review in a clean, focused modal layout. The popup shows essential details such as the user's name, email, phone number, date, and star rating along with sentiment classification.The review content is shown in a read-only field, followed by a coupon code section that can be used for promotional rewards. A "Contact via WhatsApp" button is also included to allow direct communication with the customer. The overall design is minimal and functional, maintaining the consistency of the doRev.ai platform..

# CHAPTER 6

## Technical Specification

In the development of the DoRev platform, a combination of modern technologies work together to create a robust system for automated review management, sentiment classification, and toxicity filtering. Here's how each component is utilized:

- **Frontend Technologies:**

ReactJS: ReactJS is used to develop the interactive and responsive user interface of DoRev. It enables users to easily navigate review analytics, explore sentiment trends, and manage business feedback efficiently. With component-based architecture, React ensures modular design and smooth user experience.

Tailwind CSS: Tailwind CSS handles the styling aspect of the platform, offering a clean, modern design with responsive layouts. It helps maintain visual consistency across all views, making the dashboard intuitive and accessible on different screen sizes.

- **Backend Technologies:**

TypeScript: TypeScript is employed for writing scalable and type-safe backend logic. It ensures robust handling of requests, data processing, and integration of AI models. TypeScript also enhances code maintainability, enabling reliable performance across various review management functions.

- **API Integration:**

QR Code Service: The system may use a QR code generation service to create unique codes for businesses. These codes allow customers to scan and directly submit reviews through a dedicated interface, streamlining the review collection process.

- **Algorithms and AI models:**

NLP (Sentiment Analysis): Natural Language Processing techniques are implemented to analyze the tone and context of customer reviews. This helps in identifying positive, neutral, and toxic feedback, allowing businesses to act on constructive criticism while filtering out harmful content.

- **Data and Machine Learning:**

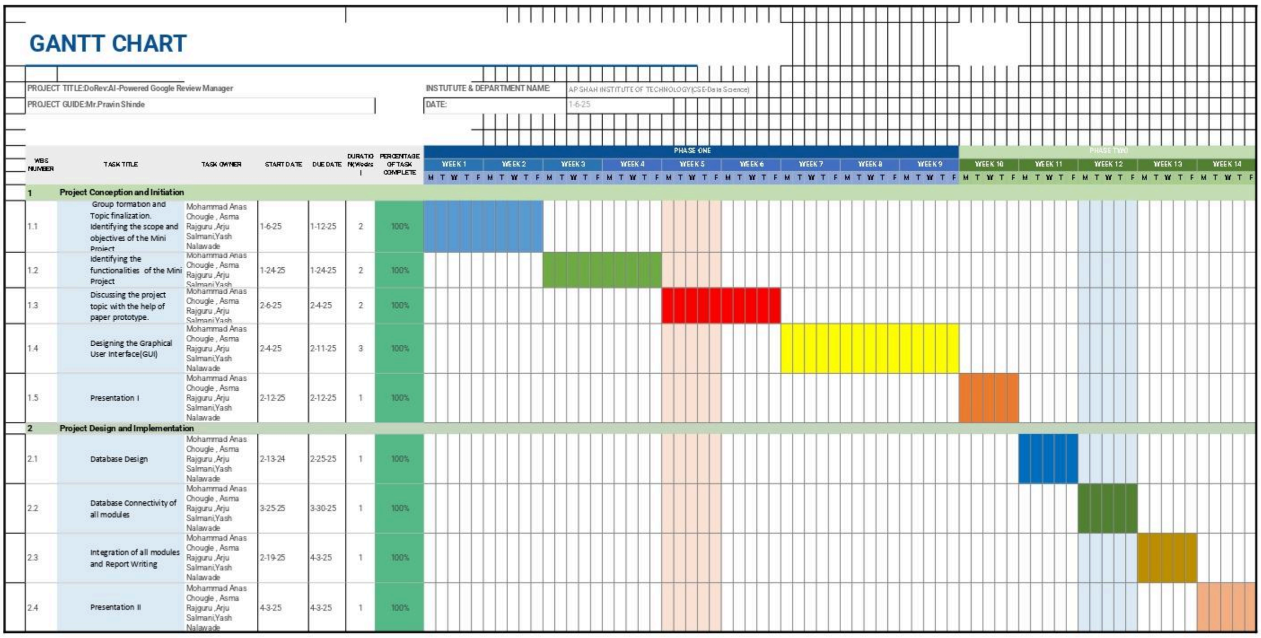
Sentiment Analysis Dataset (Kaggle): DoRev uses a publicly available sentiment analysis dataset from Kaggle to train its AI models. This dataset contains labeled review data that supports accurate sentiment classification, toxicity filtering, and insight generation. It ensures the system continuously learns and adapts to diverse review patterns, improving recommendation quality over time.

Machine Learning Algorithms:The system utilizes NLP-driven algorithms to process and classify user-generated reviews. It leverages sentiment classification techniques and toxicity detection models to evaluate review quality. These algorithms ensure that only meaningful, constructive, and positive reviews are displayed to users, enhancing feedback accuracy and brand perception.

Natural Language Processing (NLP): NLP is used to analyze customer reviews by extracting sentiment, identifying toxicity, and filtering irrelevant content. By interpreting user feedback in real time, the system helps businesses understand public opinion, detect harmful language, and highlight insightful or favorable comments for decision-making and reputation management.

# CHAPTER 7
# Project Scheduling

A Gantt chart is a visual project management tool used to plan and schedule tasks over time. It displays tasks along a timeline, showing their start and end dates, duration, and dependencies. By providing a clear view of progress, Gantt charts help teams track milestones, manage resources, and stay on schedule. This makes them essential for coordinating complex projects across various teams and stakeholders.




In Figure 7.2 Gantt Chart

**Following is the detail in Figure 7.2 of the Gantt chart:**

As in the first week of January, Mohammad Anas Chougle, Asma Rajguru, Arju Salmani, and Yash Nalawade formed a team for their mini project titled **DoRev: AI-Powered Google Review Manager**. During this phase, the team finalized the topic, scope, and objectives of the system.

By the last week of January, the team—led by Mohammad Anas Chougle and Asma Rajguru—analyzed and identified the core functional requirements of the project. In the first week of February, they collaboratively discussed the system's concept through a paper prototype, which helped solidify the workflow and structure..

By early February, Arju Salmani and Yash Nalawade began working on the Graphical User Interface (GUI) design for the platform. The first formal presentation was delivered on 12th February, where the team presented their progress and received valuable feedback from faculty members.

In the third week of February, the team, now including all members, shifted focus to Database Design to support the system's data handling. Following this, they worked on connecting the database to all modules, a task that was successfully completed by the end of March.

During the final stretch in late March to early April, the team worked on module integration and report writing. On 3rd April, they presented the final system during Presentation II, which marked the successful conclusion of their DoRev project.

# CHAPTER 8
# Result

The DoRev platform was designed to streamline review management and protect brand reputation by automating sentiment analysis and toxicity detection. The results obtained from the implementation highlight the effectiveness of the system's AI-driven approach.

- **System Overview**

The DoRev platform employs Natural Language Processing (NLP) for sentiment classification and a Deep Learning-based toxicity filter to ensure only constructive and relevant reviews are displayed.

In the figure 8.1, Sentiment Analysis method classifies customer reviews into Negative, Neutral, or Positive sentiments using Natural Language Processing (NLP). As shown in the confusion matrix, the model demonstrates high accuracy in detecting Positive reviews (96%) and Negative reviews (73%), with slight misclassifications in Neutral cases. This analysis helps businesses prioritize meaningful feedback, identify sentiment trends, and protect their brand reputation by focusing on customer satisfaction insights derived from review content.

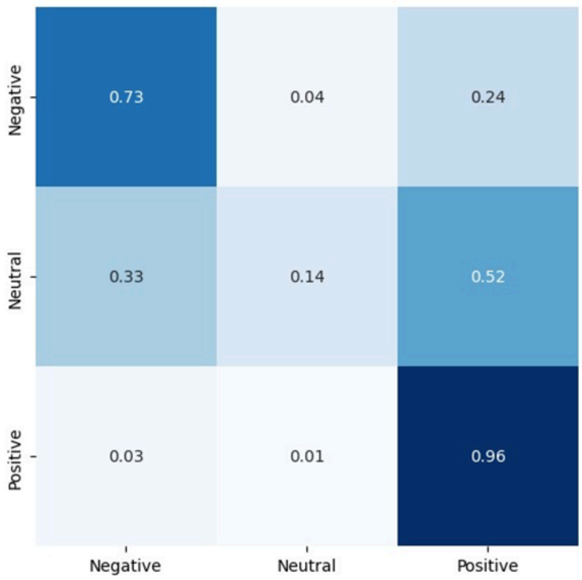



Figure 8.1 Confusion Matrix

In the figure 8.2, Content-Based Filtering method recommends reviews for publishing by analyzing specific features of each review, such as sentiment score, toxicity level, keyword presence, and relevance to the business. By focusing on the content itself rather than comparing it with other reviews, the system ensures that only meaningful and brand-safe feedback is highlighted, enhancing the quality and reliability of public reviews.

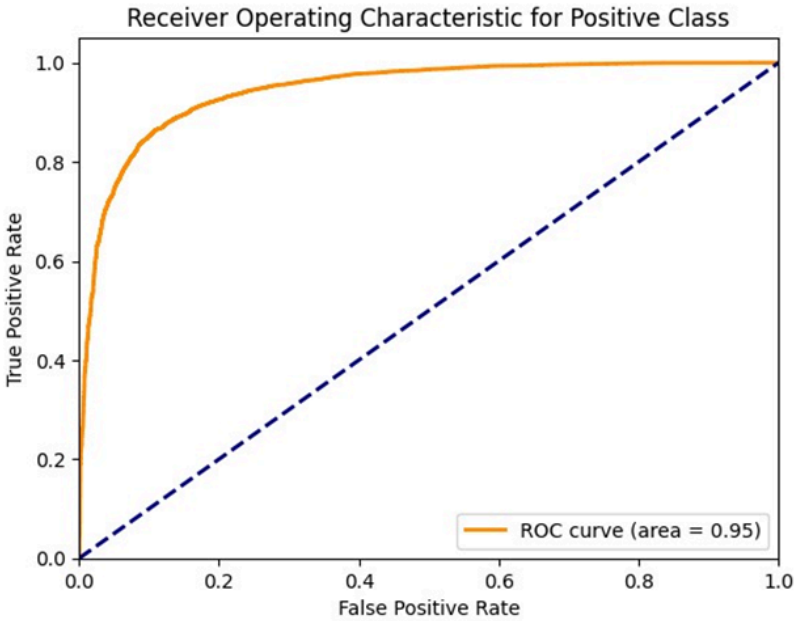


Figure 8.2 ROC Curve

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| Negative | 0.76 | 0.73 | 0.74 | 2702 |
| Neutral | 0.47 | 0.14 | 0.22 | 1021 |
| Positive | 0.89 | 0.96 | 0.93 | 10000 |
| accuracy | | | 0.86 | 13723 |
| macro avg | 0.70 | 0.61 | 0.63 | 13723 |
| weighted avg | 0.83 | 0.86 | 0.84 | 13723 |

Figure 8.3 Classification Report

Precision: The model is very precise for Positive reviews (0.89), meaning when it predicts a review as positive, it's usually correct. It's decent for Negative (0.76), but quite low for Neutral (0.47).

Recall: The model detects Positive reviews very well (0.96), fairly detects Negative (0.73), but struggles with Neutral (0.14) — meaning many neutral reviews are misclassified.

F1-score: Balanced performance metric; strongest for Positive (0.93), okay for Negative (0.74), and weak for Neutral (0.22).

Accuracy: Overall model accuracy is 0.86 (86%), which is quite strong.

Macro avg (treats all classes equally): 0.63 F1-score suggests class imbalance (which exists in your support counts).

Weighted avg (accounts for class distribution): F1-score is 0.84, reflecting the model's high performance driven by dominant Positive class.

- **Data Visualization and Model Performance:**

To evaluate the effectiveness of the system, relevant metrics were applied to each key feature.
Sentiment Analysis Performance:
Sentiment Classification Accuracy: 86.00%
Precision: 0.83
Recall: 0.86
F1-score: 0.84

- **Key insights:**

1. The sentiment analysis component shows a strong ability to correctly classify customer reviews based on emotional tone.

2. The high precision indicates that when the system labels a review as positive or negative, it is mostly accurate in doing so.

3. The recall score reflects the model's efficiency in capturing a broad range of relevant sentiments expressed in the reviews.

4. The F1-score demonstrates a well-balanced trade-off between precision and recall, ensuring overall reliability in review categorization.

**Toxicity Detection Performance:**
ROC AUC Score: 0.95

- **Key insights:**

1. The high ROC AUC score suggests excellent discriminatory power of the model to identify and filter out toxic content.

2. This indicates that the toxicity filter successfully distinguishes between constructive feedback and harmful or abusive content, ensuring the brand's digital space remains clean and professional.

- **Significance of the Results:**

The evaluation metrics confirm that DoRev is highly effective at analyzing reviews and identifying both sentiment and toxicity levels.

With strong accuracy and balanced precision-recall values, businesses can trust the system to display meaningful reviews while filtering out negative or damaging content. This leads to a healthier online presence and better-informed decision-making.

- **Challenges & Solutions:**

Although the system performs well, several development challenges were addressed:

1. Class Imbalance: Initially, there was a disproportionate number of positive reviews compared to neutral or negative ones. This was handled using oversampling techniques and balanced training sets.

2. Ambiguity in Sentiment: Some reviews had mixed tones, making them hard to classify. This was improved using more advanced NLP models and context-aware training methods.

**Future Enhancements:**

1. Multilingual Support: Expanding the model's ability to process reviews written in multiple languages to cater to a global user base.

2. Custom Business Filters: Allowing businesses to define what types of feedback they want displayed based on industry-specific needs.

3. Dashboard Expansion: Enhancing visual analytics on the dashboard to include trend timelines, comparative graphs, and keyword-based sentiment tracking.

# CHAPTER 9
# Conclusion

In conclusion, DoRev represents a significant innovation in customer feedback management by offering a robust platform that combines real-time analytics with AI-driven review filtering. By intelligently highlighting constructive and relevant reviews, applying toxicity analysis, and automating sentiment classification, DoRev empowers businesses to respond effectively to user opinions and maintain a strong online presence. The integration of sentiment datasets and chatbot models ensures businesses can efficiently interpret customer input and streamline the process of managing reviews across platforms.

The core strength of DoRev lies in its advanced NLP algorithms, which evaluate sentiment polarity, detect harmful content, and extract key insights from user feedback. The system not only filters out negative or irrelevant reviews but also ensures that valuable and positive insights are retained for business analysis. Additionally, the customizable review dashboard allows businesses to track trends, visualize customer sentiment, and adjust their strategies accordingly. This holistic review management approach ensures improved customer relations and faster, more informed decision-making.

Ultimately, DoRev's user-friendly interface makes advanced AI analytics accessible to users of all technical levels, enabling businesses to harness real-time feedback without complexity. The platform's intuitive dashboard and clear visualizations support quick interpretation of customer sentiment and review quality. Whether managing reputation, engaging with customer concerns, or optimizing service strategies, DoRev equips businesses with essential tools to operate smarter. By promoting data-driven improvement and protecting brand integrity, DoRev plays a key role in elevating customer satisfaction and advancing digital brand management.

# CHAPTER 10

## Future Scope

DoRev, with its current focus on intelligent review management, sentiment classification, and brand reputation protection, lays a strong foundation for continued development. To further expand its utility and strengthen its role as a comprehensive review analysis platform, several areas of enhancement and innovation can be pursued. These future directions aim to deliver smarter automation, deeper personalization, and ethical feedback handling, while driving valuable business insights, such as:

**1. Advanced Review Intelligence and Filtering Enhancements:**

Deep Sentiment Nuance Detection: Enhance the NLP models to detect layered sentiments within reviews, such as mixed emotions or subtle sarcasm. This can help businesses respond more accurately to complex customer feedback.

Feature-Specific Review Highlighting: Implement AI to extract and group reviews by specific product or service features (e.g., delivery time, staff behavior, product quality), allowing businesses to target key areas for improvement.

Image-Based Review Moderation: Integrate computer vision tools to analyze customer-uploaded images alongside textual reviews, offering a more comprehensive understanding of customer experiences.

**2. Real-Time Business Engagement and Response Automation:**

Smart Reply Suggestions: Introduce AI-generated response templates that adapt to the tone and content of reviews, helping businesses reply promptly and appropriately to both positive and negative feedback.

Live Feedback Tracker: Provide a dashboard module to monitor reviews in real-time with sentiment alerts, enabling businesses to take quick action on emerging reputation risks.

Automated Review Requests: Allow businesses to trigger automated follow-up messages requesting feedback from customers after transactions, increasing review volume and relevance.

**3. Community-Driven Insights and Customization:**

Trusted Reviewer Tagging: Identify and highlight reviews from verified or frequently helpful users, helping businesses and customers prioritize credible feedback.

Collaborative Analytics: Enable teams within a business to annotate reviews, add internal comments, and collaborate on customer response strategies directly within the dashboard.

Review Benchmarking: Provide comparative insights against competitor reviews in the same industry or location, helping businesses gauge their standing in the market.

**4. AI-Powered Visualization and Predictive Analytics:**

Sentiment Trend Forecasting: Use historical review data and AI models to predict upcoming sentiment trends, helping businesses proactively prepare for potential feedback shifts.

Emotion Mapping Visuals: Display heat maps and charts that reflect emotional tones across different time periods or service categories.

Toxicity Alert Overlays: Integrate visual overlays in the dashboard that flag highly toxic content and indicate urgency levels, helping moderation teams focus on critical issues efficiently.

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